

Intermediate Macro

Chapter 20: Exchange Rate Regimes

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- Following Mundell, Chapter 20 defines an **optimal currency area** as a group of countries that satisfy at least one of two conditions: similar economic shocks or high factor mobility within the group.

Flexible versus Fixed: from the Short Run to the Medium Run

20.1 The Medium Run

- When we focused on the short run in Chapter 19, we drew a sharp contrast between the behavior of an economy with a flexible exchange rate and one with a fixed exchange rate.
- It appeared to make a flexible exchange rate regime definitely more attractive than a fixed exchange rate regime.
- As we now shift the focus from the short run to the medium run, we will see that previous conclusion needs to be qualified.
- In the medium run, the economy reaches the same real exchange rate and the same level of output whether it operates under a fixed or flexible exchange rate.

Deriving the IS Relation under Fixed Exchange Rates

20.1 The Medium Run

- Start from the condition for goods market equilibrium derived in Chapter 19 (equation (19.1)):

$$Y = C(Y - T) + I(Y, r) + G + NX(Y, Y^*, \varepsilon)$$

- Next, recall the following relations:

① $r \equiv i - \pi^e$ (see Chapter 14)

② $\varepsilon = \frac{EP}{P^*}$ (see Chapter 17)

③ $E = \bar{E}$ (fixed exchange rates)

④ $i = i^*$: under fixed exchange rates and perfect capital mobility, the domestic interest rate must be equal to the foreign interest rate by uncovered interest parity condition (see Chapter 17).

The IS Relation under Fixed Exchange Rates

20.1 The Medium Run

- Using four relations, rewrite equation (19.1) as:

$$Y = C(Y - T) + I(Y, i^* - \pi^e) + G + NX(Y, Y^*, \frac{\bar{E}P}{P^*})$$

- This can be rewritten, using a more compact notation as:

$$Y = Y \left(\frac{\bar{E}P}{P^*}, G, T, i^* - \pi^e, Y^* \right) \quad (20.1)$$

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Equilibrium in the Short Run

20.1 The Medium Run

- Consider an economy where the real exchange rate is too high. As a result, the trade balance is in deficit and output is below potential.

Short Run

- Under a flexible exchange rate regime, the central bank could in principle solve the problem by decreasing the interest rate, inducing a nominal depreciation.
- Under a fixed exchange rate regime, however, the central bank cannot move the domestic interest rate. Thus, in the short run, the trade deficit remains, and the country remains in recession.

Phillips Curve Revisited

20.1 The Medium Run

- In the medium run, prices can adjust.
- We have seen that the behavior of prices is well described by the Phillips curve relation in the medium run (equation (9.3)):

$$\pi - \pi^e = \frac{\alpha}{L}(Y - Y_n)$$

- Assuming expected inflation is constant so that the Phillips curve relation is given by:

$$\pi - \bar{\pi} = \frac{\alpha}{L}(Y - Y_n) \quad (20.2)$$

- Denote the foreign inflation rate by π^* . Suppose that if output were equal to potential output, domestic and foreign inflation were equal to each other and both equal to $\bar{\pi}$.

Equilibrium in the Medium Run

20.1 The Medium Run

Medium Run

- As we are assuming that we start from a situation where output is below potential, equation (20.2) implies that $\pi < \pi^*$.
- This implies that, given the nominal exchange rate, which is fixed, the real exchange rate decreases.
- As a result, net exports increase over time, and so does output.
- In the medium run, output is back at potential, and domestic inflation is back to $\bar{\pi}$.

The Case for a Devaluation

20.1 The Medium Run

- The result that, even under fixed exchange rates, the economy can return to potential output in the medium run is important.
- But it does not eliminate the fact that the process of adjustment can be long and painful.
- If policymakers want to speed up adjustment, they can devalue the currency (decrease the level of $\bar{E}$).
- A devaluation creates a real depreciation and shifts the IS curve to the right.
- In principle, a devaluation of the right size can return the economy to its natural level of output almost immediately.

The Case against a Devaluation

20.1 The Medium Run

- In practice, however, the immediate effect of a devaluation will be to increase the price of imported goods, which has two implications.
 - ① First, it will take time for a devaluation to improve the trade balance (the J-curve effect), and
 - ② Second, devaluation will lead to an immediate increase in the cost of living (since some goods are imported), which will tend to increase wages and slow down price adjustment.
- These effects suggest that it may be difficult to determine the size of the devaluation required to restore output to its natural level.

Triggers of Exchange Rate Crises

20.2 Exchange Rate Crises under Fixed Exchange Rates

- Too much willingness on the part of government to consider devaluations actually leads to an increased likelihood of exchange rate crises.
- As soon as financial markets believe a devaluation may be coming, then maintaining the exchange rate requires an increase in the domestic interest rate.
- To see this, return to the interest parity condition derived in Chapter 17:

$$i_t = i_t^* - \frac{E_{t+1}^e - E_t}{E_t} \quad (20.3)$$

- Now suppose that there is a 75% chance the parity will be maintained and a 25% change there will be a 20% devaluation.

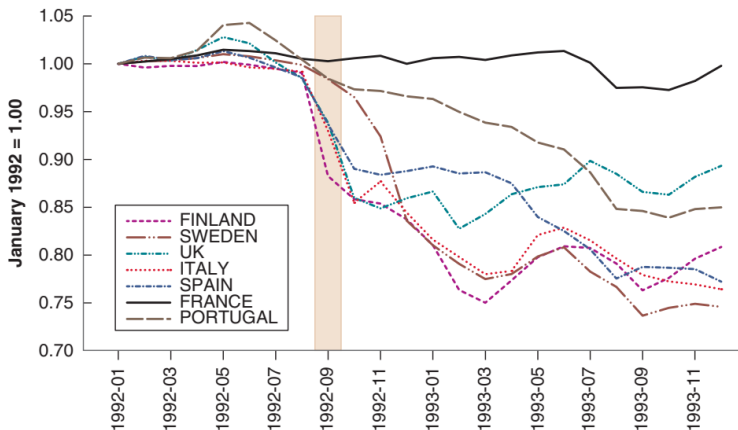
Implications of Exchange Rate Crises

20.2 Exchange Rate Crises under Fixed Exchange Rates

- This implies that, if the central bank wants to maintain the existing parity, it must now set a monthly interest rate 5% higher than before—60% higher at an annual rate.
- An interesting twist here is that a devaluation can occur even if the belief that a devaluation was coming was initially groundless.
- In other words, even if the government initially has no intention of devaluing, it might be *forced* to do so if financial markets believe that it will devalue.
- The cost of maintaining the parity would be a long period of high interest rates and a recession.
- If policymakers are unwilling to pay this cost, then a fear of devaluation can become *self-fulfilling*.

The 1992 EMS Crisis

20.2 Exchange Rate Crises under Fixed Exchange Rates



Relation between the Interest Rate and the Exchange Rate

20.3 Exchange Rate Movements under Flexible Exchange Rates

- In the model we developed in Chapter 19, there was a simple relation between the interest rate and the exchange rate: The lower the interest rate, the lower the exchange rate.
- In reality, however, the relation between the interest rate and the exchange rate is not so simple.
- To see why things are more complicated, let us return to the interest parity condition derived in Chapter 17 (equation (17.2)):

$$(1 + i_t) = (1 + i_t^*) \left(\frac{E_t}{E_{t+1}^e} \right)$$

Expectation of the Exchange Rate

20.3 Exchange Rate Movements under Flexible Exchange Rates

- Multiply both sides by E_{t+1}^e and reorganize to get

$$E_t = \frac{1 + i_t}{1 + i_t^*} E_{t+1}^e \quad (20.4)$$

- We assumed in Chapter 19 that the expected exchange rate next year E_{t+1}^e is constant. But this was a simplification.
- Write equation (20.4) for year $t + 1$ rather than for year t :

$$E_{t+1} = \frac{1 + i_{t+1}}{1 + i_{t+1}^*} E_{t+2}^e$$

- So the expectation of the exchange rate in year $t + 1$ held as of year t is given by:

$$E_{t+1}^e = \frac{1 + i_{t+1}^e}{1 + i_{t+1}^{*e}} E_{t+2}^e$$

Replacing the Expected Exchange Rates On-wards

20.3 Exchange Rate Movements under Flexible Exchange Rates

- Replacing E_{t+1}^e in equation (20.4) with the expression in the previous slide gives:

$$E_t = \frac{(1 + i_t)(1 + i_{t+1}^e)}{(1 + i_t^*)(1 + i_{t+1}^{*e})} E_{t+2}^e$$

- Continuing to solve forward in time in the same way until year $t + n$, we get:

$$E_t = \frac{(1 + i_t)(1 + i_{t+1}^e) \cdots (1 + i_{t+n}^e)}{(1 + i_t^*)(1 + i_{t+1}^{*e}) \cdots (1 + i_{t+n}^{*e})} E_{t+n+1}^e \quad (20.5)$$

Three Important Points

20.3 Exchange Rate Movements under Flexible Exchange Rates

- Equation (20.5) is sufficient to make three important points:
 - ① The level of today's exchange rate will move one-for-one with the future expected exchange rate.
 - ② Today's exchange rate will move when future expected interest rates move in either country.
 - ③ Because today's exchange rate moves with any change in expectations, the exchange rate will be *volatile*, that is, move frequently and perhaps by large amounts.
- The bottom line is that exchange rates can fluctuate greatly even in the absence of large changes in current economic variables.
- As a result, countries that operate under flexible exchange rate regimes must be prepared to accept substantial exchange rate fluctuations.

Consensus and Two Exceptions

20.4 Choosing between Exchange Rate Regimes

- Should countries choose flexible exchange rates or fixed exchange rates?
- On balance, from a macroeconomic viewpoint, it appears that flexible exchange rates are preferable to fixed exchange rates.
- There are, however, two exceptions:
 - ① First, when a group of countries is already tightly integrated, a common currency may be the right solution.
 - ② Second, when the central bank cannot be trusted to follow a responsible monetary policy under flexible exchange rates, a strong form of fixed exchange rates, such as a currency board or dollarization, may be the right solution.

Common Currency Area

20.4 Choosing between Exchange Rate Regimes

- Countries that operate under fixed exchange rates with one another are constrained to have the same interest rates.
- But how costly is that constraint?
- If the countries face roughly the same macroeconomic problems and the same shocks, they would like to have chosen similar policies in the first place.
- Forcing them to have the same monetary policy may not be much of a problem.
- This argument was first explored by Robert Mundell, who looked at the conditions under which a set of countries might want to operate under fixed exchange rates or even adopt a common currency.

Conditions for Common Currency Area

20.4 Choosing between Exchange Rate Regimes

- For countries to constitute an **optimal currency area**, they need to satisfy one of three conditions:
 - ① The countries must experience similar shocks.
 - ② Or, if the countries experience different shocks, prices and wages must be very flexible.
 - ③ Or, if the countries experience different shocks, they must have high factor mobility.
- Following Mundell's analysis, most economists believe, for example, that the common currency area composed of the 50 states of the United States is close to an optimal currency area (The third condition is largely satisfied).
- In adopting the euro, is the economic argument for the EU as compelling as it is for the United States?

Hard Pegs

20.4 Choosing between Exchange Rate Regimes

- The second case for fixed exchange rates is different from the first. It is based on the argument that there may be times when a country may want to limit its ability to monetary policy.
- If countries have a reputation for undisciplined monetary policy, international investors may fear that a flexible exchange rate system will allow too much latitude for inflationary policy.
- The country needs to convince financial investors that not only is the exchange rate fixed today, it will remain fixed in the future.
- There are two ways in which it can do so:
 - ① Make the fixed exchange rate part of a more general macroeconomic package.
 - ② Make it symbolically or technically harder to change the parity, an approach to known as a **hard peg**.

Currency Boards, and Dollarization

20.4 Choosing between Exchange Rate Regimes

- An extreme form or a hard peg is simply to replace the domestic currency with a foreign currency.
- Because the foreign currency chosen is typically the US dollar, this is known as **dollarization**.
- Few countries, however, would be willing to give up their currency and adopt the currency of another country.
- A less extreme way is to use a **currency board**. Under a currency board, a central bank stands ready to exchange foreign currency for domestic currency at the official exchange rate set by the government.
- Furthermore, the central bank cannot engage in open market operations.
- The best-known example of a currency board is that adopted by Argentina in 1991.